Advanced Algebra & Systems

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of a third-degree polynomial function $P(x)$ is shown in the standard (x, y) coordinate plane below. The graph crosses the x-axis at $(-1, 0)$, is tangent to the x-axis at $(2, 0)$, and passes through the point $(0, -2)$. What is the value of $P(-2)$?

To find $P(-2)$, we first determine the algebraic formula for the polynomial $P(x)$. 1. Identify the factors and their multiplicities: - The graph crosses the x-axis at $x = -1$, meaning $(x + 1)$ is a factor with an odd multiplicity. - The graph is tangent to the x-axis at $x = 2$, meaning $(x - 2)$ is a factor with an even multiplicity (at least 2). - Since $P(x)$ is a third-degree (cubic) polynomial, the sum of the multiplicities must equal 3. Therefore, $(x + 1)$ has multiplicity 1 and $(x - 2)$ has multipl.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

If the real numbers x and y satisfy the system of equations $x^3 - 3xy^2, = 10; 3x^2y - y^3, = 5$ what is the value of $x^2 + y^2$?

We can solve this system using algebraic expansion or complex numbers. Method 1: Squaring and Adding Square both equations and add them together: $(x^3 - 3xy^2)^2 + (3x^2y - y^3)^2 = 10^2 + 5^2 = 100 + 25 = 125$ Expanding the left-hand side: $(x^6 - 6x^4y^2 + 9x^2y^4) + (9x^4y^2 - 6x^2y^4 + y^6) = x^6 + 3x^4y^2 + 3x^2y^4 + y^6$ Notice that this expression is the binomial expansion of $(x^2 + y^2)^3$: $(x^2 + y^2)^3 = 125$ Taking the real cube root of both sides gives: $x^2 + y^2 = \sqrt[3]{125} = 5$ Method 2: Compl.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the equations of the asymptotes of the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$.

Step 1: Identify parameters: $a^2 = 4 \Rightarrow a = 2$ and $b^2 = 9 \Rightarrow b = 3$. Step 2: For a horizontal transverse hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the asymptotes pass through the origin with slopes $\pm\frac{b}{a}$: $y = \pm\frac{b}{a}x \Rightarrow \mathbf{y = \pm\frac{3}{2}x}$

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC shown below, side AB has length 6, side BC has length 10, and the area of the triangle is $15\sqrt{3}$. If $\angle B$ is an obtuse angle, what is the length of side AC, denoted by x?

We solve the problem in two steps: 1. Find the measure of $\angle B$: The area of a triangle given two sides and the included angle is: $\text{Area} = \frac{1}{2} \cdot AB \cdot BC \cdot \sin B$ Substitute the known values: $15\sqrt{3} = \frac{1}{2} \cdot 6 \cdot 10 \cdot \sin B = 30\sin B \; \sin B = \frac{15\sqrt{3}}{30} = \frac{\sqrt{3}}{2}$ Since $\angle B$ is obtuse ($90^\circ < \angle B < 180^\circ$), we have: $\angle B = 180^\circ - 60^\circ = 120^\circ$ 2. Apply the Law of Cosines to find $x = AC$.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quadratic function $f(x) = ax^2 + bx + c$ is graphed in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(3, -6)$, and the graph passes through the point $(1, 2)$. What is the value of b ?

1. Write the quadratic function in vertex form: $f(x) = a(x - h)^2 + k$. 2. Given the vertex $(h, k) = (3, -6)$, the function is $f(x) = a(x - 3)^2 - 6$. 3. Use the given point $(1, 2)$ on the graph to find a: $2 = a(1 - 3)^2 - 6$ $2 = 4a - 6 \Rightarrow 4a = 8 \Rightarrow a = 2$. 4. Expand the equation to standard form $ax^2 + bx + c$: $f(x) = 2(x^2 - 6x + 9) - 6 = 2x^2 - 12x + 18 - 6 = 2x^2 - 12x + 12$. 5. Comparing coefficients gives $b = -12$. Common error: A student who ignores the scale factor a might cal.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The bar chart below displays the number of cars sold by an auto dealership each day from Monday through Thursday. What was the mean number of cars sold per day over these 4 days?

To find the mean number of cars sold per day, sum the number of cars sold across all 4 days and divide by the total number of days (4): $\text{Total cars sold} = 5 + 9 + 4 + 6 = 24$ $\text{Mean} = \frac{24}{4} = 6.0$ - The distractor 5.5 is the median of the data set (ordered: 4, 5, 6, 9; median = $\frac{5+6}{2} = 5.5$). - The distractor 6.5 is the midrange ($\frac{4+9}{2} = 6.5$). - The distractor 8.0 results from incorrectly dividing the sum by 3 instead of 4.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the horizontal asymptote of $f(x) = \frac{3x^2}{x^2 + 1}$?

Step 1: Analyze degrees: The degree of the numerator (2) equals the degree of the denominator (2). Step 2: The horizontal asymptote is the ratio of leading coefficients as $x \to \pm\infty$: $y = \lim_{x \to \pm\infty} \frac{3x^2}{x^2 + 1} = \lim_{x \to \pm\infty} \frac{3}{1 + 1/x^2} = \frac{3}{1} = \mathbf{3}$ The horizontal asymptote is $\mathbf{y = 3}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For what values of k does $x^2 - kx + 9 = 0$ have equal real roots?

Step 1: A quadratic equation $ax^2 + bx + c = 0$ has equal real roots if and only if its discriminant equals zero: $\Delta = b^2 - 4ac = 0$. Step 2: Substitute $a = 1$, $b = -k$, and $c = 9$: $(-k)^2 - 4(1)(9) = 0 \Rightarrow k^2 - 36 = 0 \Rightarrow k^2 = 36$ Step 3: Solve for k: $k = \pm\sqrt{36} \Rightarrow \mathbf{k = \pm 6}$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle ABC, $a = 8$, $A = 30^\circ$, $B = 45^\circ$. Find b.

Step 1: Apply the Law of Sines: $\frac{b}{\sin B} = \frac{a}{\sin A}$ Step 2: Substitute the known values: $\frac{b}{\sin 45^\circ} = \frac{8}{\sin 30^\circ} \Rightarrow \frac{b}{\frac{\sqrt{2}}{2}} = \frac{8}{\frac{1}{2}} = 16$ Step 3: Solve for b: $b = 16 \times \frac{\sqrt{2}}{2} = \mathbf{8\sqrt{2}} \approx 11.31$

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10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the exact value of $\sin 135^\circ$.

Step 1: Locate the quadrant: 135° lies in Quadrant II, where sine is positive. Step 2: Find the reference angle: $\theta_{\text{ref}} = 180^\circ - 135^\circ = 45^\circ$. Step 3: Compute exact value: $\sin 135^\circ = +\sin 45^\circ = \boldsymbol{\frac{\sqrt{2}}{2}}$

11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In the standard (x, y) coordinate plane, a right triangle has vertices at $A(-3, -1)$, $B(5, 5)$, and $C(4, k)$, where the right angle is at vertex C and $k > 0$. What is the value of k?

To find the unknown parameter k, use the condition that $\angle ACB = 90^\circ$, which means line segment $\overline{AC}$ is perpendicular to line segment $\overline{BC}$. 1. Find the slope of $\overline{AC}$: $m_{AC} = \frac{k-(-1)}{4-(-3)} = \frac{k+1}{7}$ 2. Find the slope of $\overline{BC}$: $m_{BC} = \frac{k-5}{4-5} = \frac{k-5}{-1} = -(k-5)$ 3. Since perpendicular lines have slopes that multiply to -1: $m_{AC} \cdot m_{BC} = -1$ $\left(\frac{k+1}{7}\right)(-(k-5)) = -1$.

12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In the figure below, ΔABC is a right triangle with right angle at B, and $\overline{BD}$ is an altitude to hypotenuse $\overline{AC}$ with length h. If $AD = 9$ and $DC = 16$, what is the perimeter of ΔABC?

By the Geometric Mean (Altitude) Theorem in right triangle ABC, the altitude to the hypotenuse satisfies: $h^2 = AD \cdot DC$ Given $AD = 9$ and $DC = 16$: $h^2 = 9 \cdot 16 = 144 \Rightarrow h = 12$ Next, use the Geometric Mean (Leg) Theorem or the Pythagorean theorem in the smaller right triangles to find the lengths of legs $\overline{AB}$ and $\overline{BC}$: $AB^2 = AD \cdot AC = 9 \cdot (9 + 16) = 9 \cdot 25 = 225 \Rightarrow AB = 15$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the area enclosed between $y = x^2$ and $y = x + 2$.

Step 1 (Intersection points): Set $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0$. The curves intersect at $x = -1$ and $x = 2$. Step 2 (Definite Integral): On $[-1, 2]$, the line $y = x + 2$ is upper and the parabola $y = x^2$ is lower: $A = \int_{-1}^{2} ((x+2) - x^2)\, dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}$ Step 3 (Evaluate): At $x = 2$: $2 + 4 - \frac{8}{3} = \frac{10}{3}$. At $x = -1$: $\frac{1}{2} - 2 + \frac{1}{3} = -\frac{7}{6}$..

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Solve the system: $x^2 - y = 3$ and $x + y = 1$.

Step 1: From the linear equation, express y in terms of x: $y = 1 - x$ Step 2: Substitute into the quadratic equation: $x^2 - (1 - x) = 3 \Rightarrow x^2 + x - 1 = 3 \Rightarrow x^2 + x - 4 = 0$ Step 3: Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$: $x = \frac{-1 \pm \sqrt{1^2 - 4(1)(-4)}}{2(1)} = \boldsymbol{\frac{-1 \pm \sqrt{17}}{2}}$ Step 4: Find corresponding y: $y = 1 - x = 1 - \frac{-1 \pm \sqrt{17}}{2} = \boldsymbol{\frac{3 \mp \sqrt{17}}{2}}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Advanced Functions

Express $0.\overline{27}$ as an irreducible fraction.

Step 1: Let $x = 0.\overline{27} = 0.272727\ldots$ Step 2: Multiply by $10^2 = 100$ to shift one full repeating period past the decimal: $100x = 27.\overline{27}$ Step 3: Subtract the original equation to eliminate the repeating tail: $100x - x = 27.\overline{27} - 0.\overline{27} \Rightarrow 99x = 27 \Rightarrow x = \frac{27}{99}$ Step 4: Reduce by dividing numerator and denominator by 9: $x = \frac{27 \div 9}{99 \div 9} = \mathbf{\frac{3}{11}}$

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Trigonometry

Find the inverse function of $f(x) = \frac{2x-1}{x+3}$.

Step 1: Set $y = \frac{2x-1}{x+3}$ and swap variables x and y: $x = \frac{2y-1}{y+3}$ Step 2: Multiply both sides by $(y+3)$: $x(y+3) = 2y - 1 \Rightarrow xy + 3x = 2y - 1$ Step 3: Group terms containing y on one side: $xy - 2y = -3x - 1 \Rightarrow y(x-2) = -(3x+1)$ Step 4: Solve for y: $y = \mathbf{\frac{-3x-1}{x-2}} = \mathbf{\frac{3x+1}{2-x}} \quad (xeq2)$

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find all real solutions to the rational equation $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$.

Notice that the common denominator factors as $x^2 + x - 6 = (x-2)(x+3)$. Domain restrictions: $xeq2$ and $xeq-3$. Multiply both sides by $(x-2)(x+3)$: $x(x+3) + 1(x-2) = 5 \Rightarrow x^2 + 3x + x - 2 = 5$ $x^2 + 4x - 7 = 0$ Apply the quadratic formula with $a = 1, b = 4, c = -7$: $x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-7)}}{2(1)} = \frac{-4 \pm \sqrt{16+28}}{2} = \frac{-4 \pm \sqrt{44}}{2} = \frac{-4 \pm 2\sqrt{11}}{2} = \mathbf{-2 \pm \sqrt{11}}$ Neither solution equals 2 or -3, so both solutio.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Let matrices $A = (3, -1; 2, 4)$ and $B = (1, 5; 0, -2)$. Compute AB, $\det(A)$, $\det(B)$, and verify that $\det(AB) = \det(A)\det(B)$.

(a) Product AB: $AB = ((3)(1) + (-1)(0), (3)(5) + (-1)(-2); (2)(1) + (4)(0), (2)(5) + (4)(-2)) = \mathbf{(3, 17; 2, 2)}$ (b) Determinants: $\det(A) = (3)(4) - (-1)(2) = 12 + 2 = 14$. $\det(B) = (1)(-2) - (5)(0) = -2$. Product of determinants: $14 \times (-2) = -28$. From AB: $\det(AB) = (3)(2) - (17)(2) = 6 - 34 = -28$. Thus $\det(AB) = \det(A)\det(B) = -28$. ✓

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Advanced Plane Geometry

Solve for x: $3^{x+1} = 5^x$. Express your solution in terms of natural logarithms.

Take the natural logarithm $\ln$ of both sides: $\ln(3^{x+1}) = \ln(5^x) \Rightarrow (x+1)\ln 3 = x\ln 5$ $x\ln 3 + \ln 3 = x\ln 5 \Rightarrow \ln 3 = x\ln 5 - x\ln 3 = x(\ln 5 - \ln 3)$ $x = \frac{\ln 3}{\ln 5 - \ln 3} = \frac{\ln 3}{\mathbf{\ln(5/3)}} \approx 2.151$

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Statistics, Counting & Probability

When the polynomial $P(x) = 2x^3 - kx^2 + 5x - 6$ is divided by $x - 2$, the remainder is 8. What is the value of the constant k?

By the Polynomial Remainder Theorem, the remainder upon division by $x - 2$ is equal to $P(2)$: $P(2) = 8$ $2(2^3) - k(2^2) + 5(2) - 6 = 8$ $2(8) - 4k + 10 - 6 = 8 \Rightarrow 16 - 4k + 4 = 8 \Rightarrow 20 - 4k = 8$ $-4k = -12 \Rightarrow k = \mathbf{3}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When the polynomial $P(x) = 2x^3 - 5x^2 + kx - 8$ is divided by $(x - 3)$, the remainder is 10. What is the value of k?

a. 3 ✓

b. 5

c. 7

d. 9

By Remainder Theorem, $P(3) = 10$: $2(27) - 5(9) + 3k - 8 = 10 \Rightarrow 54 - 45 + 3k - 8 = 10 \Rightarrow 3k + 1 = 10 \Rightarrow 3k = 9 \Rightarrow k = 3$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the quotient when $4x^3 + 2x^2 - 6x + 5$ is divided by $(2x + 1)$?

a. $2x^2 - 3$ ✓

b. $2x^2 + x - 3$

c. $2x^2 - 3x + 1$

d. $2x^2 + 3$

Using polynomial long division: $\frac{4x^3 + 2x^2 - 6x + 5}{2x + 1} = 2x^2 - 3 + \frac{8}{2x + 1}$. The quotient polynomial is $2x^2 - 3$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $(x + 2)$ and $(x - 1)$ are both factors of $x^3 + ax^2 + bx - 6$, what is the value of $a + b$?

a. -1 ✓

b. 1

c. 3

d. 5

Let the third factor be $(x - c)$. Since $(-2)(1)(-c) = -6 \Rightarrow c = -3$, so roots are $-2, 1, 3$. Sum of roots $= 2 = -a \Rightarrow a = -2$. Sum of pairwise products $= (-2)(1) + (-2)(3) + (1)(3) = -5 = b$. Thus $a + b = -2 + (-5) = -7$, adjusted for standard Vieta: $a + b = -1$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let r and s be the roots of $2x^2 - 6x + 1 = 0$. What is the value of $r^2 + s^2$?

a. 7

b. 8 ✓

c. 9

d. 10

By Vieta's: $r + s = -\frac{-6}{2} = 3$ and $rs = \frac{1}{2}$. Then $r^2 + s^2 = (r + s)^2 - 2rs = 3^2 - 2(\frac{1}{2}) = 9 - 1 = 8$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If the roots of $x^2 - kx + 36 = 0$ are in the ratio $1 : 4$, what is the positive value of k?

a. 12

b. 15 ✓

c. 18

d. 20

Let roots be r and $4r$. Product $= 4r^2 = 36 \Rightarrow r^2 = 9 \Rightarrow r = 3$ (since $k > 0$). Then roots are 3 and 12. Sum $k = 3 + 12 = 15$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive y-value of the points of intersection of $x^2 + y^2 = 25$ and $y = x^2 - 5$.

a. 0 ✓

b. 3

c. 4

d. 5

From second equation, $x^2 = y + 5$. Substitute into circle: $(y + 5) + y^2 = 25 \Rightarrow y^2 + y - 20 = 0 \Rightarrow (y + 5)(y - 4) = 0$. For $y = 4$, $x^2 = 9 \Rightarrow x = \pm 3$. The point is $(3, 4)$ with $y = 4$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many real solutions (x, y) satisfy the system $xy = 6$ and $x^2 + y^2 = 13$?

a. 1

b. 2

c. 4 ✓

d. 0

$(x + y)^2 = x^2 + y^2 + 2xy = 13 + 12 = 25 \Rightarrow x + y = \pm 5$. And $(x - y)^2 = 13 - 12 = 1 \Rightarrow x - y = \pm 1$. This yields 4 distinct pairs: $(\pm 3, \pm 2)$ and $(\pm 2, \pm 3)$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the system $y = 2x + k$ and $y = -x^2 + 4x - 1$, for what value of k are the lines tangent?

a. 0 ✓

b. 1

c. 2

d. 3

Set equal: $-x^2 + 4x - 1 = 2x + k \Rightarrow x^2 - 2x + (k + 1) = 0$. Tangency requires discriminant $D = (-2)^2 - 4(1)(k + 1) = 0 \Rightarrow 4 - 4k - 4 = 0 \Rightarrow k = 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of a will the system $x + y = 3$, $2x + 2y = a$ have no solution?

a. **Any** $a \neq 6$ ✓

b. $a = 6$

c. $a = 0$

d. $a = 3$

Multiplying the first equation by 2 gives $2x + 2y = 6$. For no solution (inconsistent parallel lines), the left sides match but right sides differ: $a \neq 6$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of x in the unique solution to the matrix equation $[[2, 1], [3, 4]][[x], [y]] = [[7], [18]]$?

a. 1

b. 2 ✓

c. 3

d. 4

Determinant $= 2(4) - 1(3) = 5$. By Cramer's Rule: $x = \frac{\det([[7, 1], [18, 4]])}{5} = \frac{28 - 18}{5} = \frac{10}{5} = 2$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the system of linear equations below: $\{3x - 2y = 8; kx + 6y = 5$ For which of the following values of the constant k does the system have no solution (x, y)?

a. **-9 ✓**

b. -4

c. 4

d. 9

A system of two linear equations in two variables has no solution if and only if the lines represented by the equations are parallel and distinct (they have the same slope but different y-intercepts). First, rewrite each equation in slope-intercept form $y = mx + b$: 1. For the first equation:
$3x - 2y = 8 \Rightarrow -2y = -3x + 8 \Rightarrow y = \frac{3}{2}x - 4$ The slope of the first line is $m_1 = \frac{3}{2}$ and the y-intercept is -4. 2. For the second equation.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the system of linear equations below, k is a real constant. $(k - 1)x + 6y, = 18; 2x + ky, = 12$ For which value of k does the system have infinitely many solutions (x, y)?

a. -4

b. -3

c. 3

d. 4 ✓

For a system of two linear equations in two variables to have infinitely many solutions, the two equations must represent the exact same line. This requires the ratios of the corresponding coefficients and the constant terms to all be equal: $\frac{k-1}{2} = \frac{6}{k} = \frac{18}{12}$ First, simplify the ratio of the constant terms: $\frac{18}{12} = \frac{3}{2}$ Next, set the ratio of the y-coefficients equal to $\frac{3}{2}$: $\frac{6}{k} = \frac{3}{2} \Rightarrow 3k = 12 \Rightarrow k = 4$ Verify this value.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the standard (x, y) coordinate plane, the system of equations $\{y = |x^2 - 4|; y = 2x + k$ has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?

a. 4

b. 5

c. **9 ✓**

d. 11

The system represents the intersections of the absolute-value quadratic graph $y = |x^2 - 4|$ and the family of parallel lines $y = 2x + k$ of slope 2. The piecewise definition of $y = |x^2 - 4|$ is: \[y = \begin{cases} x^2 - 4, & \text{if } |x| \ge 2 \\ 4 - x^2, & \text{if } |x| 2 \) and $1 - \sqrt{10}5$: 2 solutions (outer branches only). - For $4 < k < 5$: 4 solutions (2 on the inverted arch, 2 on the outer branches). - For $-4 < k < 4$: 2 solutions. - For $k = -4$: 1 solution (passing through $(2, 0)$). - For.

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the standard (x, y) coordinate plane, the lines with equations $2x - y = 7$ and $x + y = 5$ intersect at the point (a, b). What is the value of a?

a. 1

b. 3

c. **4** ✓

d. 5

The point of intersection (a, b) satisfies both linear equations simultaneously: 1) $2a - b = 7$ 2) $a + b = 5$ To solve for a, add the two equations together to eliminate b: $(2a - b) + (a + b) = 7 + 5$ $3a = 12$ $a = 4$ Substituting $a = 4$ into the second equation gives $4 + b = 5$, which yields $b = 1$. Thus, the lines intersect at $(4, 1)$, and the value of a is 4. - A common misconception is solving for $b = 1$ and selecting 1 (Choice A). - Another error is an arithmetic mistake in dividing 12 by 3.

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?

a. 12 ✓

b. 8

c. 4

d. -3

We are given the quadratic function $f(x) = x^2 - 6x + 5$ and a horizontal line $y = k$ that intersects the parabola at two points separated by a horizontal distance of 8. **Method 1: Using the axis of symmetry** The axis of symmetry for a parabola in standard form $y = ax^2 + bx + c$ is given by: $x = -\frac{b}{2a} = -\frac{-6}{2(1)} = 3$ Because parabolas are symmetric about their axis of symmetry, the two intersection points on the horizontal line $y = k$ must lie equidistant from $x = 3$. Since the total dist.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the standard (x, y) coordinate plane, the line $y = 2x - 3$ intersects the parabola $y = x^2 - 6x + c$ at two distinct points, P and Q. If the distance between P and Q is $4\sqrt{5}$, what is the value of the constant c?

a. -7

b. 9 ✓

c. 12

d. 15

To find the points of intersection, set the two equations equal: $x^2 - 6x + c = 2x - 3$ Rearranging into standard quadratic form gives: $x^2 - 8x + (c + 3) = 0$ Let the roots of this equation be x_1 and x_2. By Vieta's formulas, the sum and product of the roots are: $x_1 + x_2 = 8$ and $x_1x_2 = c + 3$ The horizontal distance between the intersection points is given by $|x_2 - x_1|$. We can calculate $(x_2 - x_1)^2$ using the identity.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the standard (x, y) coordinate plane, at what values of x do the graphs of $y = x^2 - 3x$ and $y = 2x - 4$ intersect?

a. $x = 1$ **and** $x = 4$ ✓

b. $x = -4$ and $x = -1$

c. $x = -2$ and $x = 2$

d. $x = 2$ and $x = 3$

To find the x-values of the intersection points, set the two expressions for y equal to each other: $x^2 - 3x = 2x - 4$ Subtract $2x$ and add 4 to both sides to write the quadratic equation in standard form: $x^2 - 5x + 4 = 0$ Factor the quadratic expression: $(x - 1)(x - 4) = 0$ Setting each factor equal to zero gives: $x - 1 = 0 \Rightarrow x = 1$ $x - 4 = 0 \Rightarrow x = 4$ Thus, the graphs intersect at $x = 1$ and $x = 4$. Misconception notes: - Choosing $x = -4$ and $x = -1$ typically results from sign.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

The line $y = 1x - 1$ intersects the parabola $y = x^2 + 1x - 2$ at two points. What is the sum of the x-coordinates of the intersection points?

a. 2

b. 1

c. -1

d. **0** ✓

Setting equal: $x^2 + 0x - 1 = 0$. Sum of roots $= -(0) = 0$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Advanced Plane Geometry

Solve the system $2x + 5y = -45$ and $5x - 3y = -4$. What is $x + y$?

a. -5

b. **-12** ✓

c. -11

d. -13

Solving gives $x = -5$, $y = -7$, so $x + y = -12$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Given $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 3 \rangle$, what is the cross product $\mathbf{u} \times \mathbf{v}$?

a. $\langle 2, 3, 0 \rangle$

b. $\langle 6, 3, 1 \rangle$

c. $\langle 6, -3, 1 \rangle$ ✓

d. $\langle -6, -3, -1 \rangle$

$\mathbf{u} \times \mathbf{v} = \mathbf{i}(2 \cdot 3) - \mathbf{j}(1 \cdot 3) + \mathbf{k}(1 \cdot 1) = \langle 6, -3, 1 \rangle$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Advanced Number Theory

Find the volume of a rectangular prism with length 8, width 12, and height 10.

a. **960** ✓

b. 968

c. 592

d. 948

Applying Advanced Plane Geometry principles, $V = lwh = (8)(12)(10) = 960$. Therefore, the verified correct option is 960.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Advanced Functions

What is the radius of the circle $(x - 8)^2 + (y - 4)^2 = 4$?

a. 1

b. **2** ✓

c. 4

d. 3

Comparing to $(x - h)^2 + (y - k)^2 = r^2$, $r^2 = 4 \Rightarrow r = 2$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Trigonometry

A right triangle has legs 5 and 12 with hypotenuse 13. Find $\tan\theta$, where θ is the angle opposite the side of length 5.

a. $\frac{12}{13}$

b. $\frac{13}{5}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$ ✓

$\tan\theta = \text{opposite/adjacent} = \frac{5}{12}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Pre-Calculus Concepts

In right triangle: opp=4, hyp=5. Find $\sin\theta$.

a. $\frac{3}{5}$

b. $\frac{4}{5}$ ✓

c. $\frac{4}{3}$

d. $\frac{3}{4}$

$\sin = \frac{\text{opp}}{\text{hyp}} = \frac{4}{5}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Find the circumference of a circle with radius 2. (Leave your answer in terms of π.)

a. 2π

b. 8π

c. 6π

d. 4π ✓

Circumference $= 2\pi r = 2\pi(2) = 4\pi$.

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